

Computer Vision Take-Home Lab

Intensity Transformations — Changing What the Pixel Sees

Learning Objectives

- Explain pixel intensity.
- Apply basic intensity transformations using Python.
- Observe brightness and contrast changes.
- Experiment with negative transformation, thresholding, and gamma correction.
- Compare results and explain how pixel values changed.

Key idea: Geometric transformations change **where pixels are located**, while intensity transformations change the **values of the pixels themselves**.

Important: Do not simply run the code. Change parameter values, observe the results, and explain what happened.

Preparation — Google Colab

Create a Google Colab notebook and upload your own photograph.

```
import cv2
import numpy as np
```

```
import matplotlib.pyplot as plt
from google.colab import files

uploaded = files.upload()
filename = list(uploaded.keys())[0]

image = cv2.imread(filename)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8,6))
plt.imshow(image)
plt.axis("off")
plt.show()
```

Convert it to grayscale:

```
gray = cv2.cvtColor(image, cv2.COLOR_RGB2GRAY)

plt.imshow(gray, cmap="gray")
plt.axis("off")
plt.show()
```

Describe your original image:

The image I used was a picture taken of me during our football match of this year's BU Olympics

Original image dimensions:

720x1280

1. Meet Your Pixels

```
print("Minimum intensity:", gray.min())
print("Maximum intensity:", gray.max())
print("Mean intensity:", gray.mean())
```

```
y = 100
x = 100
```

```
print("Pixel intensity:", gray[y, x])
```

1. What is the minimum possible intensity in an 8-bit grayscale image?
2. What is the maximum?
3. What would a pixel value of 0 look like?
4. What would a pixel value of 255 look like?
5. Find a dark and bright area. What do you expect their pixel values to be?

2. 255

3. Black, since it has no intensity

4. White, since it has high intensity

5. The dark areas have lower values, while the lighter areas have higher values

2. Brightness — Make It Lighter or Darker

```
brighter = cv2.add(gray, 50)
darker = cv2.subtract(gray, 50)

plt.figure(figsize=(15,5))
plt.subplot(1,3,1); plt.imshow(gray, cmap="gray"); plt.title("Original");
plt.axis("off")
plt.subplot(1,3,2); plt.imshow(brighter, cmap="gray"); plt.title("Brighter");
plt.axis("off")
plt.subplot(1,3,3); plt.imshow(darker, cmap="gray"); plt.title("Darker");
plt.axis("off")
plt.show()
```

Try 20, 80, and 120 instead of 50.

1. What happens as the value increases?
2. What happens when the value becomes very large?
3. Why can't an 8-bit pixel become brighter than 255?
4. Does brightness change object positions or pixel values?

1. As the values increase, the picture becomes darker

2. When the i use a really large value like 200, the whole picture becomes black

3. Because an 8-bits means 2^8 , which results to 256, and ince 0 is included, it cant hold values greater than 255

3. Contrast — Stretch the Difference

```
contrast = cv2.convertScaleAbs(gray, alpha=1.8, beta=0)

plt.figure(figsize=(12,5))
plt.subplot(1,2,1); plt.imshow(gray, cmap="gray"); plt.title("Original");
plt.axis("off")
plt.subplot(1,2,2); plt.imshow(contrast, cmap="gray"); plt.title("Higher Contrast");
plt.axis("off")
plt.show()
```

Try alpha values of 0.5, 1.0, 1.8, and 2.5.

Alpha	What happened?
0.5	Barely no difference in the picture
1.0	Slight brightness change
1.8	Picture becomes significantly brighter
2.5	Picture becomes nearly totally white

What is the difference between brightness and contrast adjustment?

Brightness adjustment changes the overall lightness or darkness of an image, while contrast adjustment changes the difference between dark and bright pixels.

4. Negative Transformation

```
negative = 255 - gray

plt.figure(figsize=(12,5))
plt.subplot(1,2,1); plt.imshow(gray, cmap="gray"); plt.title("Original");
plt.axis("off")
plt.subplot(1,2,2); plt.imshow(negative, cmap="gray"); plt.title("Negative");
plt.axis("off")
plt.show()

y = 100
x = 100
original_value = gray[y, x]
negative_value = negative[y, x]
print("Original:", original_value)
print("Negative:", negative_value)
```

If the original pixel is **37**, what should its negative value be?

Why does $s = 255 - r$ produce a negative-looking image?

The formula reverses the intensity values, white (0) becomes black (255), Dark gray (37) - Bright gray (218), etc. Therefore, dark areas become bright and bright areas become dark

5. Thresholding — Black or White?

```
threshold_value = 128

_, binary = cv2.threshold(
    gray, threshold_value, 255, cv2.THRESH_BINARY
)

plt.figure(figsize=(12,5))
plt.subplot(1,2,1); plt.imshow(gray, cmap="gray"); plt.title("Grayscale");
plt.axis("off")
```

```
plt.subplot(1,2,2); plt.imshow(binary, cmap="gray"); plt.title("Binary");
plt.axis("off")
plt.show()
```

Try thresholds of 50, 100, 128, 180, and 220.

1. What happens when the threshold is very low?
2. What happens when it is very high?
3. Which threshold works best for your image?
4. Why could thresholding be useful for object detection or segmentation?

2. Most pixels will be below the threshold, so they become black. Only the brightest areas will remain white.

3. 128, it clearly separates me out from the background

4. Thresholding separates pixels into foreground and background. This makes it easier

6. Gamma Correction

```
gamma = 2.0
```

```
normalized = gray / 255.0
gamma_corrected = np.power(normalized, gamma)
gamma_corrected = np.uint8(gamma_corrected * 255)
```

```
plt.figure(figsize=(12,5))
plt.subplot(1,2,1); plt.imshow(gray, cmap="gray"); plt.title("Original");
plt.axis("off")
plt.subplot(1,2,2); plt.imshow(gamma_corrected, cmap="gray"); plt.title("Gamma
Corrected"); plt.axis("off")
plt.show()
```

Try gamma = 0.5, 1.0, 2.0, and 3.0.

Which gamma made the image brighter?

0.5

Which made it darker?

3.0

Is gamma correction simply adding the same value to every pixel?

No gamma correction uses a different formula to determine the value, therefore, different pixel values are changed by different amounts.

What makes gamma correction different from simple brightness adjustment?

Simple brightness adjustment typically adds or subtracts a fixed amount from every pixel, gamma correction is nonlinear, meaning it changes different intensity ranges differently

7. Intensity Transformation Challenge

Choose one photograph and apply **at least four different intensity transformations**.

Example: Brightness → Contrast → Negative → Threshold

Image	Transformation	Observation
Original	None	The
Image 1	Brightness	I could change the brightness or darkness of the image. My image was quite dark since it was taken at night, so i made
Image 2	Contrast	When I increased the contrast, it made the darker areas darker and the lighter

Image 3	Negative	The darker areas became light, and the light areas became dark
Image 4	Threshold	I am separated from the background of the image, and I can clearly see the outlines

Which transformation produced the most interesting result? Why?

Threshold. I could see how this could be useful in computer vision task such as object detection and OCR

Final Reflection — Geometric vs. Intensity

1. Pixels 2. Brightness 3. Contrast 4. Negative 5. Threshold 6. Gamma
7. Challenge Reflection Submission

Situation B: Every pixel becomes 50 intensity levels brighter.

Which situation is geometric and which is intensity transformation? Explain.

Situation A is a geometric transformation because the image is moved 100 pixels to the right, changing the positions of the objects. Situation B is an intensity transformation because the pixel values become brighter while their positions remain the same.

If object positions stay the same but pixel values change, what kind of transformation is this?

This is an intensity transformation because only the pixel intensity values are changed. The objects remain in their original positions.

In 5–7 sentences, explain intensity transformation to someone who has never studied Computer Vision.

Intensity transformation is a process of changing the brightness or intensity of pixels in an image. It does not move the objects or change their positions in the image. Instead, it changes how light or dark the pixels appear. For example, increasing brightness makes the pixels lighter, while decreasing brightness makes them darker. Contrast adjustment changes the difference between dark and bright areas. Negative transformation reverses the brightness values while thresholding changes pixels into

Why might intensity transformation be useful before object detection, segmentation, or feature extraction?

Intensity transformation can make objects and important features easier to see in an image. It can improve object detection, segmentation, and feature extraction by making differences between objects and their backgrounds more noticeable.

Submission Checklist

- ☒ I used an appropriate image.
- ☒ I ran and modified the Python code in Google Colab.
- ☒ I completed the investigation questions.
- ☒ I completed the transformation challenge.
- ☒ I included screenshots of important outputs.
- ☒ I included my Google Colab notebook/code.
- ☒ I explained what I observed, not only the output.

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